CAEM: Confidence-Aware Episodic Memory with Self-Improvement

Md. Aksan Gony Alif | Maliha Binte Shamim | Md. Mahmudur Rahman
Department of Computer Science & Engineering, BRAC University

Headline result (vs. matched zero-shot baseline B1)

Training-panel architectural-contribution slice (CAEM C2 vs. B1):

$\boxed{\Delta\text{EM} = +20.3\%\text{ rel.}}$ $\boxed{\Delta\text{CHM} = -29.4\%\text{ rel.}}$

Pooled training-panel EM lifts from 0.483 at B1 to 0.581 at CAEM C2; pooled training-panel CHM falls from 0.143 to 0.101. CAEM beats every one of the eight external baselines on both axes at both reference cycles (8/8 wins-against). The retrieval-utility classifier ablation lifts **CommonsenseQA EM by** +14.0 **pp** and **TruthfulQA EM by** +10.0 **pp** by re-routing misconception-bait queries away from the retrieval-augmented tier.

Problem: hallucination as a deployment-safety issue

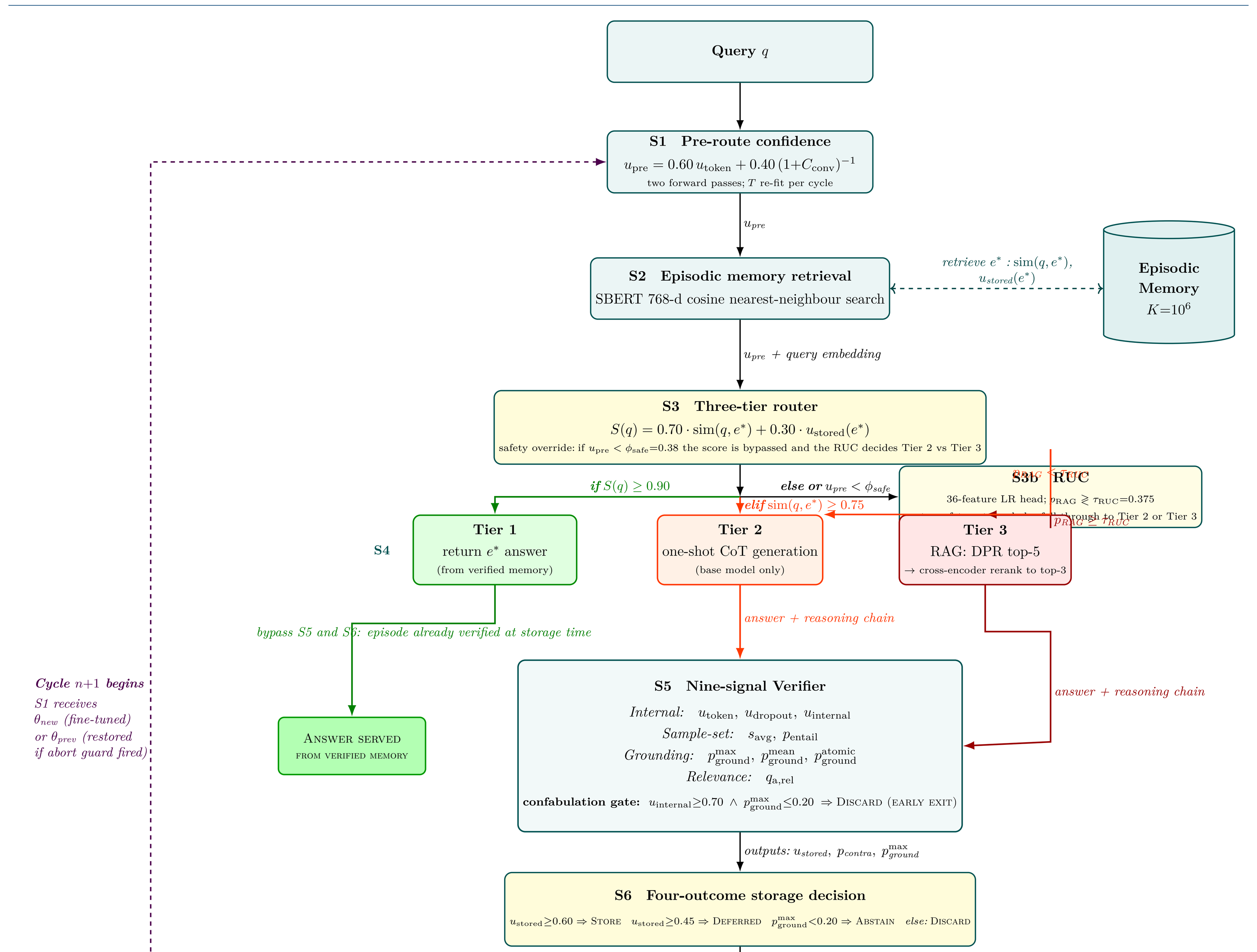
Modern instruction-tuned language models generate confident-sounding answers at deployment-relevant rates that exceed their actual accuracy by ten or more percentage points [1,2]. Single-signal correctness predictors achieve AUROC scores between 0.50 and 0.60, marginally above chance [3]. The TruthfulQA benchmark documents an inverse-scaling pattern under which larger models are *more* prone to echo human falsehoods [4]. Clinical-question studies report hallucination rates of 69 – 77% on frontier models [5].

The three structural shortcomings

- **Statelessness:** no representation of past verified answers persists across sessions.
- **Overconfidence:** single-signal correctness predictors are near random.
- **Immutable capability:** naive online fine-tuning produces catastrophic forgetting.

CAEM is an architectural intervention that closes all three jointly.

The architecture: eight-stage pipeline + cycle-boundary loop



Nine-signal verifier + calibrated probability composite

Four signal families feed the per-benchmark calibrated probability composite:

- **Internal calibration:** u_{token}, $u_{dropout}$, $u_{internal}$
- **Sample-set agreement:** s_{avg}, p_{entail}
- **External grounding:** p^{max}_{ground}, p^{mean}_{ground}, length-gated p^{atomic}_{ground}
- **Question-answer relevance:** $q_{a,rel}$

Calibrated probability composite

Per-signal isotonic regression maps each raw signal to a calibrated probability; per-benchmark child curves are shrunk toward the pooled fit through a James-Stein-style prior at $\alpha = 0.6$; an L_2-regularised logistic boost layer ($C = 0.01$) aggregates the per-signal log-odds into $u_{stored} \in [0, 1]$.

Storage gate (fixed threshold)

$$u_{stored} \geq 0.60 \to \textsc{Store} \quad u_{stored} \geq 0.45 \to \textsc{Defer} \quad \text{else} \to \textsc{Abstain} / \textsc{Discard}$$

Thresholds are frozen across cycles; only the composite refits per cycle. Replaces an earlier split-conformal gate rejected on the registered calibration-to-evaluation distribution-shift evidence.

Self-improvement loop with asymmetric-rollback retention guard

At every committed cycle boundary:

1. **Retroverify** every stored episode under the **locked verifier ensemble** (MiniCheck + frozen Qwen judge), re-scored against the updated generator's outputs and the cycle-boundary-refit composite. Prune entries that fall below $\tau_{retro} = 0.50$.
2. **Reconsider** every entry in the deferred buffer (promote, drop, keep) under TTL = 2 cycles.
3. **Consolidate** near-duplicate-meaning clusters under a cosine-similarity floor of 0.92.
4. **Fine-tune** the LoRA adapter on the strict training pool ($\tau_{train} = 0.70$); three epochs, effective batch 16.
5. **Probe** the registered multi-modal retention panel (MMLU, TQA-test, CSQA-test). If the worst-probe ratio drops below $\rho_{min} = 0.93$, the adapter rolls back to the previous committed checkpoint while **the memory store is preserved across the rollback** (the asymmetric-rollback contract).

Crucially, only the *generator* (advancing via SIL) and the *composite* (refitting per cycle) are cycle-indexed; the verifier ensemble is held fixed by design, so the per-cycle cal-fold class separation is a joint observable of the SIL-advanced generator viewed through that fixed lens.

Cycle-boundary trajectory: SIL + memory + commit pattern

Cycle	SIL n	Final loss	Worst probe	Commit	Memory pool
C1	107	0.258	1.006	✓	2 818
C2 ⋆	1 474	0.171	0.949	✓	5 151
C3	2 966	0.109	0.976	✓	7 477
C4	3 624	0.070	**0.886**	× **ABORT**	9 961
C5	5 006	0.091	0.958	✓	**12 175**

The realised commit pattern $(1, 1, 1, 0, 1)$ yields commit fraction $\widehat{\pi}_{commit} = 4/5 = 0.80$, strictly in the open interval $(0, 1)$. The memory pool grows monotonically from 431 cold-start entries to 12 175 at the cycle-five close **even across the cycle-four retention abort**, because the stream-chunk path executes independently of the retention guard under the asymmetric-rollback contract: deployment-time cost amortisation (memory tier growth) decouples from training-time parametric progression (adapter commits). Final-cycle losses decline monotonically across the four uninterrupted fine-tunes ($0.258 \to 0.171 \to 0.109 \to 0.070$), the empirical receipt that SIL makes coherent gradient progress on the verified pool.

Per-cycle pooled trajectory (5-benchmark evaluation panel)

Cycle	Pooled EM	Pooled CHM	MMLU	TQA-test	CSQA-test
B1 (zero-shot ref)	0.507	0.120	—	—	—
C0	0.541	0.135	—	—	—
C1	0.543	0.112	1.041	1.013	1.006
C2 ⋆	**0.544**	**0.107**	0.975	0.949	0.982
C3	0.532	0.115	0.992	0.987	0.976
C4 **abort**	0.532	0.115	1.018	**0.886**	0.982
C5 (final)	0.517	0.109	0.999	1.013	0.958

The composite hallucination metric reduces −19% relative across the trajectory and attains its minimum at C2. C4 fires the retention guard on TQA-test = 0.886 < ρ_{min} = 0.93; C5 recovers to 0.958. Pool purity holds in the band [0.732, 0.746] on the cal fold across all committed cycles, clearing the registered 0.64 floor by 9 – 11 pp.

External eight-baseline panel (matched protocol)

	System	EM	CHM	ΔEM vs C2	ΔCHM vs C2
⋆	**CAEM C2 (best)**	**0.544**	**0.107**	—	—
	CAEM C5 (final)	0.517	0.109	−0.027	+0.002
B1	zero-shot	0.507	0.120	+7.4%	−11.0%
B2	chain-of-thought	0.500	0.123	+8.8%	−12.7%
B3	DPR-RAG	0.458	0.148	+18.8%	−27.7%
B4	CoT + DPR-RAG	0.461	0.142	+17.9%	−24.6%
B5	five-shot CoT	0.405	0.136	+34.2%	−21.4%
B6	vanilla fine-tune (C5)	0.325	0.144	+67.2%	−25.5%
B7	FLARE	0.292	0.153	+86.3%	−29.9%
B8	semantic entropy	0.505	0.120	+7.7%	−10.5%

Wins-against: CAEM C2 beats 8/8 **baselines; CAEM C5 beats** 8/8**.** CHM is post-hoc rescored through the locked CAEM verifier for every row, so the column is apples-to-apples. TruthfulQA EM uses the Haiku-judged correctness rule for every row including the abstention-licensed B8.

Retrieval-utility-classifier ablation (C3 close)

Benchmark	EM off	EM on	ΔEM (pp)	T3 share off	T3 share on	Δ
FEVER	0.503	0.503	0.0	92.3%	70.7%	−21.7
TriviaQA	0.437	0.463	+2.7	95.7%	68.0%	−27.7
CommonsenseQA	0.603	**0.743**	**+14.0**	57.7%	20.3%	−37.3
StrategyQA	0.617	0.640	+2.3	100.0%	76.7%	−23.3
TruthfulQA	0.210	**0.310**	**+10.0**	99.0%	40.0%	**−59.0**
Pooled	**0.474**	**0.532**	+5.8	**88.9**%	**55.1**%	**−33.8**

The asymmetric-rescue pattern (bold cells): the largest EM gains land on CommonsenseQA and TruthfulQA, the two benchmarks where retrieval was actively hurting in the classifier-off arm. FEVER EM stays flat: the classifier correctly preserves the retrieval-helpful regime on a textbook fact-verification benchmark.

Code, data, and references

Code: `github.com/aksaN000/caem-thesis` **Artefacts (Google Drive):** `drive.google.com/drive/folders/1xErhsRDYe_q0OZgUBn0WXK81eHc5jZCa` **Passage index (HF, private):** `aksaN000/caem-passage-index-21m`

[1] Guo et al., *ICML 2017.* [2] Kadavath et al., *arXiv 2022.* [3] Xiong et al., *ICLR 2024.* [4] Lin et al., *ACL 2022.* [5] Alkaissi & McFarlane, *Cureus 2023.* [6] Thorne et al., *NAACL 2018.* [7] Joshi et al., *ACL 2017.* [8] Talmor et al., *NAACL 2019.* [9] Geva et al., *TACL 2021.* [10] Hu et al., *ICLR 2022* (LoRA). [11] Tang et al., *2024* (MiniCheck). [12] Cherian et al., *2024* (boost layer).